

NETWORK SWITCHING LAB

VLAN Configuration, VTP & Inter-VLAN Routing

Simulated on EVE-NG | Cisco IOS Switches | 6-Switch Topology

Lab Title	Network Switching: VLANs, Trunking & Inter-VLAN Routing
Platform	EVE-NG (Emulated Virtual Environment Next Generation)
Devices Used	Core Switch (L3) + Switch 1–5 (Access/Distribution) + 16 PCs + Windows Server
VLANs Configured	VLAN 10 (PC), VLAN 20 (PC), VLAN 30 (PC), VLAN 40 (PC), VLAN 50 (Server)
VTP Domain	corvit Core = Server Access Switches = Client
Protocols	IEEE 802.1Q Trunking, VTP v1, Inter-VLAN Routing via SVIs
Institute	Corvit Institute Networking Department

1. Lab Objectives

This lab demonstrates the configuration of a multi-switch network using VLANs, IEEE 802.1Q trunk ports, VTP (VLAN Trunking Protocol), and Layer 3 Inter-VLAN routing. The topology simulates a real enterprise access-distribution-core hierarchy on EVE-NG.

- Configure trunk links between Core switch and all access switches (Switch 1–5)
- Create VLANs 10, 20, 30, 40, and 50 on the Core switch (VTP Server)
- Propagate VLAN database to all access switches automatically via VTP
- Assign end-host ports as access ports in their respective VLANs
- Configure Switch Virtual Interfaces (SVIs) on Core for Inter-VLAN routing
- Verify trunking, VLAN distribution, routing, and end-to-end connectivity

2. Network Topology

The topology consists of one Core (Layer 3) switch connected to four access switches (Switch 1–4) and one uplink switch (Switch 5) that aggregates the Windows Server. All uplinks from access to core are 802.1Q trunk links. PCs are connected to access switches through access-mode ports.

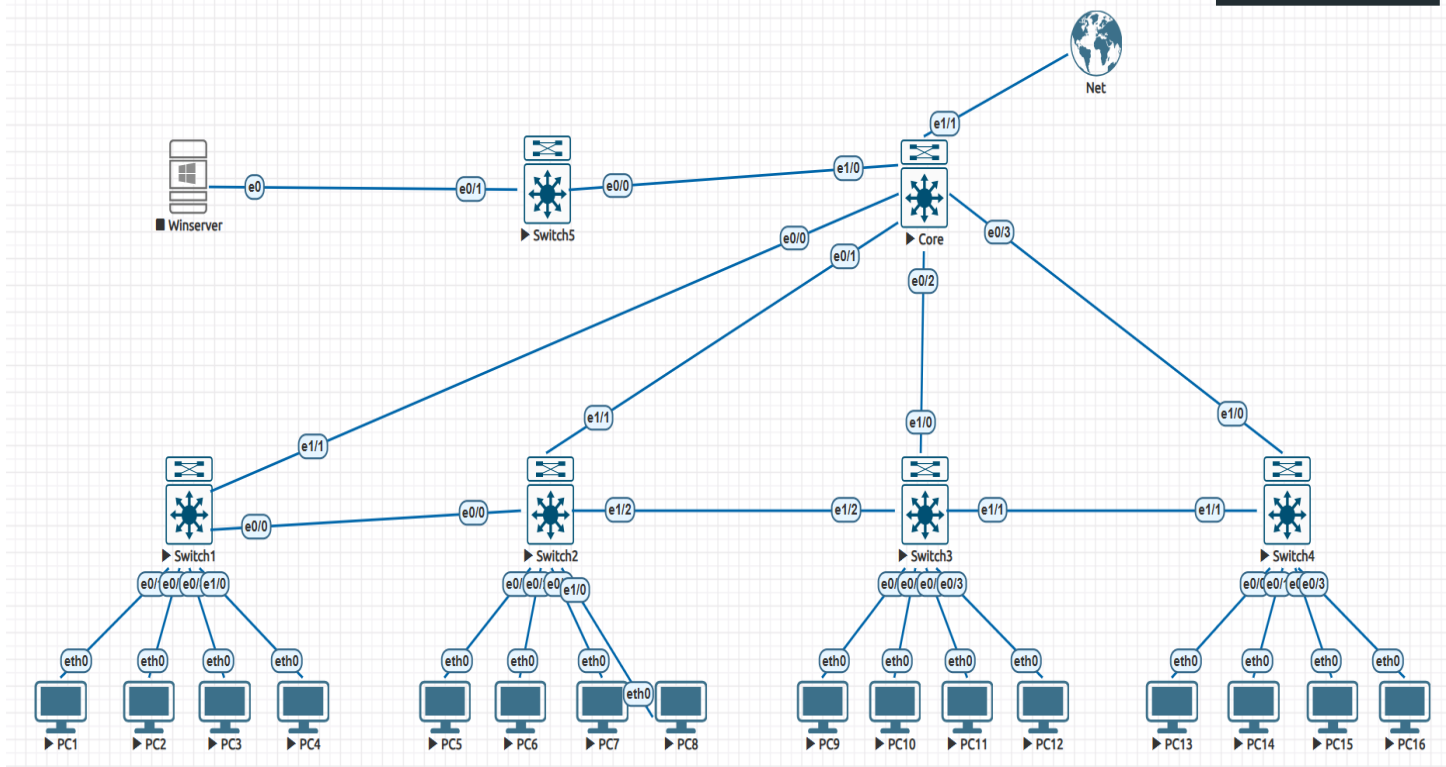


Figure: EVE-NG Topology Core + 4 Access Switches + Switch5 (Server) + 16 PCs

Switch	Role	Connected To	Interfaces Used
Core (L3)	Distribution / Routing	Switch1–5, Internet	e0/0–e0/3, e1/0–1
Switch 1	Access	Core, PC1–PC4	e0/0–e1/0
Switch 2	Access	Core, PC5–PC8	e0/0–e1/0
Switch 3	Access	Core, PC9–PC12	e0/0–3
Switch 4	Access	Core, PC13–PC16	e0/0–3
Switch 5	Server Aggregation	Core, Windows Server	e0/0–1

3. VLAN Design & IP Addressing

Five VLANs were designed to logically segment the network. Each VLAN maps to a /24 subnet. The Core switch acts as the gateway (SVI) for every VLAN, enabling inter-VLAN communication.

VLAN ID	Name	Subnet	Gateway (SVI)	Purpose
10	PC_VLAN_10	192.168.10.0/24	192.168.10.1	End-user PCs (Group A)
20	PC_VLAN_20	192.168.20.0/24	192.168.20.1	End-user PCs (Group B)

VLAN ID	Name	Subnet	Gateway (SVI)	Purpose
30	PC_VLAN_30	192.168.30.0/24	192.168.30.1	End-user PCs (Group C)
40	PC_VLAN_40	192.168.40.0/24	192.168.40.1	End-user PCs (Group D)
50	Server_VLAN_50	192.168.50.0/24	192.168.50.1	Windows Server

3.1 End-Device IP Address Table

Device	VLAN	IP Address	Subnet Mask	Default Gateway
PC1	10	192.168.10.11	255.255.255.0	192.168.10.1
PC2	20	192.168.20.11	255.255.255.0	192.168.20.1
PC3	30	192.168.30.11	255.255.255.0	192.168.30.1
PC4	40	192.168.40.11	255.255.255.0	192.168.40.1
PC5	10	192.168.10.12	255.255.255.0	192.168.10.1
PC6	20	192.168.20.12	255.255.255.0	192.168.20.1
PC7	30	192.168.30.12	255.255.255.0	192.168.30.1
PC8	40	192.168.40.12	255.255.255.0	192.168.40.1
PC9	10	192.168.10.13	255.255.255.0	192.168.10.1
PC10	20	192.168.20.13	255.255.255.0	192.168.20.1
PC11	30	192.168.30.13	255.255.255.0	192.168.30.1
PC12	40	192.168.40.13	255.255.255.0	192.168.40.1
PC13	10	192.168.10.14	255.255.255.0	192.168.10.1
PC14	20	192.168.20.14	255.255.255.0	192.168.20.1
PC15	30	192.168.30.14	255.255.255.0	192.168.30.1
PC16	40	192.168.40.14	255.255.255.0	192.168.40.1
WinServer	50	192.168.50.10	255.255.255.0	192.168.50.1

4. Step-by-Step Configuration

4.1 Configure Trunk Ports on All Switches

All uplinks between access switches and the Core must be set to trunk mode using 802.1Q encapsulation. Apply these commands on each switch's uplink interface:

```
! On each Switch - uplink interface toward Core
enable
configure terminal
interface ethernet 0/0          ! uplink port toward Core
    switchport trunk encapsulation dot1q
    switchport mode trunk
    no shutdown
exit
```

4.2 Create VLANs on Core Switch (VTP Server)

Create all five VLANs on the Core switch and assign descriptive names. VTP will propagate these to client switches automatically.

```
! On CORE switch
enable
configure terminal
vlan 10
    name PC_VLAN_10
vlan 20
    name PC_VLAN_20
vlan 30
    name PC_VLAN_30
vlan 40
    name PC_VLAN_40
vlan 50
    name Server_VLAN_50
exit
```

4.3 Configure VTP (VLAN Trunking Protocol)

Set the Core as VTP Server and all access switches as VTP Clients in the 'corvit' domain. This ensures VLANs are synchronized automatically across all switches.

```
! On CORE switch (VTP Server)
vtp domain corvit
vtp mode server
```

```
! On each Access Switch (Switch1 - Switch5)
vtp domain corvit
vtp mode client
```

4.4 Configure Access Ports on Access Switches

Assign each PC-facing port as an access port in the appropriate VLAN. Repeat per-interface for all PCs. Example shown for Switch 1:

```
! Switch 1 - Access port examples
interface e0/1                                ! PC1 - VLAN 10
    switchport mode access
    switchport access vlan 10
    no shutdown
interface e0/2                                ! PC2 - VLAN 20
    switchport mode access
    switchport access vlan 20
    no shutdown
interface e0/3                                ! PC3 - VLAN 30
    switchport mode access
    switchport access vlan 30
    no shutdown
interface e1/0                                ! PC4 - VLAN 40
    switchport mode access
    switchport access vlan 40
    no shutdown
```

4.5 Configure SVIs for Inter-VLAN Routing on Core

Enable IP routing on the Core (Layer 3) switch and create a Switch Virtual Interface (SVI) for each VLAN. These SVIs act as the default gateways for all PCs in each VLAN.

```
! On CORE switch
ip routing
interface vlan 10
    ip address 192.168.10.1 255.255.255.0
    no shutdown
interface vlan 20
    ip address 192.168.20.1 255.255.255.0
    no shutdown
interface vlan 30
    ip address 192.168.30.1 255.255.255.0
    no shutdown
interface vlan 40
    ip address 192.168.40.1 255.255.255.0
    no shutdown
interface vlan 50
    ip address 192.168.50.1 255.255.255.0
    no shutdown
exit
```

5. Verification & Results

After completing all configuration steps, the following verification commands were executed on the Core switch to confirm that VTP synchronization, VLAN distribution, trunk links, routing, and end-to-end connectivity are all functioning correctly.

5.1 VTP Status Verification — show vtp status

Confirms Core is operating as VTP Server in the 'corvit' domain with 10 VLANs and Configuration Revision 5.

```
VTP Version capable      : 1 to 3
VTP version running     : 1
VTP Domain Name         : corvit
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID               : aabb.cc80.6000
Configuration last modified by 0.0.0.0 at 5-20-26 14:57:30
Local updater ID is 192.168.10.1 on interface Vl10 (lowest numbered
VLAN interface found)

Feature VLAN:
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 10
Configuration Revision   : 5
MD5 digest               : 0xBA 0x58 0x1D 0x25 0xBE 0xD1 0
x2E 0x98
                          0x36 0x7E 0xCB 0xD4 0xC4 0xF2 0
x4A 0x17
CORE#
```

Figure: Core Switch show vtp status output

5.2 VLAN Database show vlan brief

Confirms all five VLANs (10, 20, 30, 40, 50) are present and active on the Core switch.

```
CORE#sh vlan br

VLAN Name                Status    Ports
-----
1    default                active    Et1/1, Et1/2, Et1/3
10   PC_VLAN_10              active
20   PC_VLAN_20              active
30   PC_VLAN_30              active
40   PC_VLAN_40              active
50   Server_VLAN_50          active
1002 fddi-default             act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
CORE#
```

Figure: Core Switch show vlan brief output

5.3 Trunk Link Status show interfaces trunk

Confirms all five uplink ports (Et0/0 – Et1/0) are operating in trunk mode with 802.1Q encapsulation.

```
CORE#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vl
an				
Et0/0	on	802.1q	trunking	1
Et0/1	on	802.1q	trunking	1
Et0/2	on	802.1q	trunking	1
Et0/3	on	802.1q	trunking	1
Et1/0	on	802.1q	trunking	1

Figure: Core Switch show interfaces trunk output

5.4 SVI / Routing Status show ip interface brief

Confirms all five SVI interfaces (Vlan10 – Vlan50) are UP/UP with their gateway IP addresses correctly assigned.

```
CORE#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	unassigned	YES	unset	up	up
Ethernet0/1	unassigned	YES	unset	up	up
Ethernet0/2	unassigned	YES	unset	up	up
Ethernet0/3	unassigned	YES	unset	up	up
Ethernet1/0	unassigned	YES	unset	up	up
Ethernet1/1	unassigned	YES	unset	up	up
Ethernet1/2	unassigned	YES	unset	up	up
Ethernet1/3	unassigned	YES	unset	up	up
Vlan10	192.168.10.1	YES	manual	up	up
Vlan20	192.168.20.1	YES	manual	up	up
Vlan30	192.168.30.1	YES	manual	up	up
Vlan40	192.168.40.1	YES	manual	up	up
Vlan50	192.168.50.1	YES	manual	up	up

Figure: Core Switch show ip interface brief output

5.5 End-to-End Connectivity VPCS Ping Test

PC1 (VLAN 10, 192.168.10.11) successfully pings PC13 (VLAN 10, 192.168.10.14) on Switch 4, confirming intra-VLAN routing across trunk links.

```
VPCS> show ip
```

NAME : VPCS[1]
 IP/MASK : 192.168.10.11/24
 GATEWAY : 192.168.10.1
 DNS :
 MAC : 00:50:79:66:68:08
 LPORT : 20000
 RHOST:PORT : 127.0.0.1:30000
 MTU : 1500

```
VPCS> ping 192.168.10.14
```

84 bytes from 192.168.10.14 icmp_seq=1 ttl=64 time=52.387 ms
 84 bytes from 192.168.10.14 icmp_seq=2 ttl=64 time=183.959 ms
 84 bytes from 192.168.10.14 icmp_seq=3 ttl=64 time=164.151 ms
 84 bytes from 192.168.10.14 icmp_seq=4 ttl=64 time=27.882 ms
 84 bytes from 192.168.10.14 icmp_seq=5 ttl=64 time=323.691 ms

Figure: VPCS PC1 pinging PC13 (192.168.10.14) 5/5 packets received

6. Command Reference Summary

Command	Purpose
<code>switchport trunk encapsulation dot1q</code>	Set trunk encapsulation to IEEE 802.1Q
<code>switchport mode trunk</code>	Force port into permanent trunk mode
<code>vlan 10 / name PC_VLAN_10</code>	Create VLAN and assign descriptive name
<code>vtp domain corvit</code>	Set VTP administrative domain
<code>vtp mode server / client</code>	Set VTP operating mode on each switch
<code>ip routing</code>	Enable Layer 3 routing on Core switch
<code>interface vlan 10</code>	Enter SVI configuration for VLAN 10
<code>ip address 192.168.10.1 255.255.255.0</code>	Assign gateway IP to SVI
<code>no shutdown</code>	Bring interface/SVI up
<code>show vtp status</code>	Verify VTP domain, mode, revision, VLAN count
<code>show vlan brief</code>	Display all VLANs and their status
<code>show interfaces trunk</code>	Verify trunk links and encapsulation
<code>show ip interface brief</code>	Check SVI IP addresses and UP/UP state
<code>write / wr</code>	Save running configuration to NVRAM

7. Conclusion

This lab successfully demonstrated a complete enterprise switching architecture using Cisco IOS switches simulated in EVE-NG. The following outcomes were achieved:

- **All trunk links between Core and access switches are operational with 802.1Q encapsulation**
- **Five VLANs were created on Core and automatically propagated to all access switches via VTP**
- **Each PC was assigned to its designated VLAN through access-mode port configuration**
- **SVIs on the Core switch provide inter-VLAN routing, enabling cross-VLAN communication**
- **End-to-end connectivity was verified via ping tests confirming correct routing and forwarding**